

SSM Schrodinger + Auto + Evolution Assurance

A concise three-way integration plan for the current product, research-enabling Auto capabilities, and the proposed reliability research extensions.

Core proposition: SSM specifies and deterministically generates the product; Auto supplies reproducibility, trace, contract, and artifact evidence; the research layer determines whether behaviour across releases is ordinary variation, intended evolution, regression, or inconclusive.

Purpose

Create a production-aligned research setup without changing the identity of SSM as a semantic application compiler. The integration should use Auto as a trust and observation substrate, then add a statistically controlled evolution-assurance layer above individual generation runs.

Lane	What exists or is added	Primary output
1. Current SSM + adopted Auto trust functions	Intent-to-app compilation, capability honesty, deterministic target generation, bounded SML repair, manifests, contracts, eval records, provenance, assumptions, unsupported-feature reports and evidence bundles.	A valid generated product with run-level evidence.
2. Additional Auto integration	Trace/replay, typed task graph, determinism census, behavioural contracts, differential verification, content-addressed eval runs and signed registry records.	Immutable observations suitable for repeated-run comparison.
3. Research assay and extensions	Matched repeated runs, paired/sequential tests, change-intent envelopes, release verdicts, hidden-slice detection and stage attribution.	A release-level evolution-assurance decision.

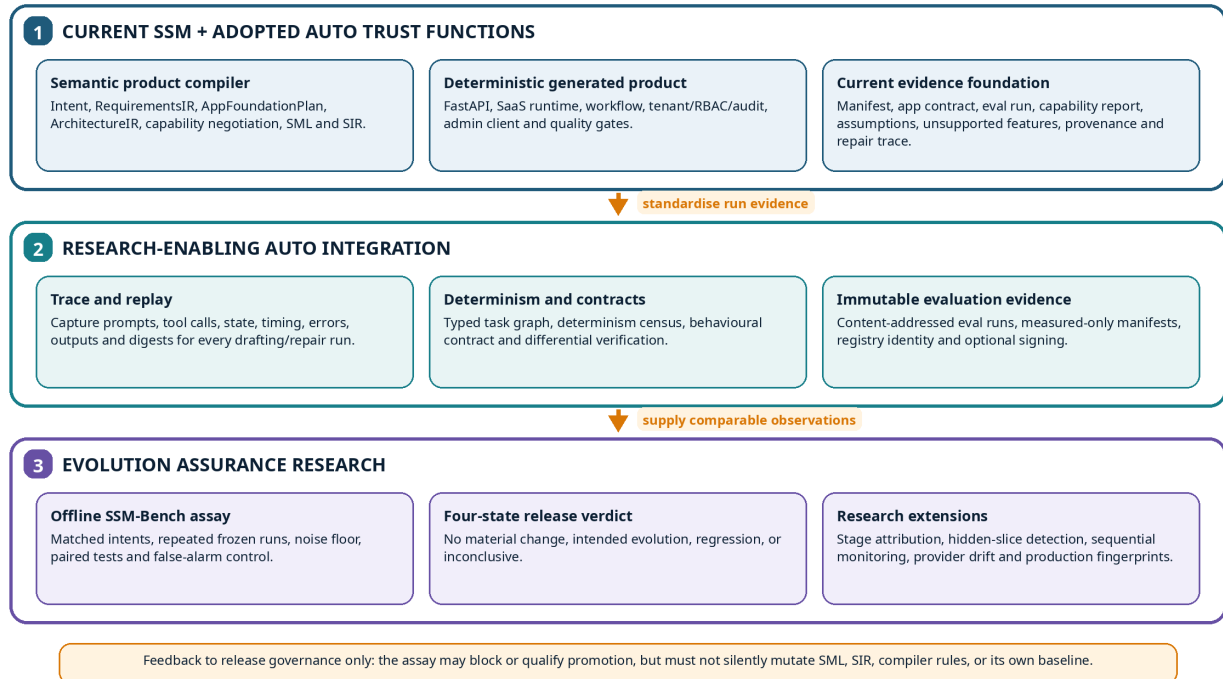
Recommended product boundary: keep compilation, observation, and statistical judgement as separate control planes. A statistical monitor may block or qualify a release, but must not silently rewrite compiler semantics or adapt its own baseline.

Visual reference architecture

The three lanes are sequential but retain explicit feedback loops. Research findings create release-gate verdicts and new telemetry requirements; they do not directly mutate SML, SIR, or deterministic code generation.

Three-way integration reference

Product compiler -> reproducibility substrate -> evolution assurance



Decision flow: individual SSM runs produce verified artifacts and evidence -> Auto-derived observation records make runs comparable -> the research assay compares releases and returns an explicit four-state verdict.

Four-state verdict

Verdict	Meaning	Release action
NO MATERIAL CHANGE	Observed differences remain inside the calibrated noise floor.	Proceed, subject to normal gates.
INTENDED EVOLUTION	A statistically real change remains inside an approved change-intent envelope.	Proceed with the trade-off recorded in the evidence bundle.
REGRESSION	A statistically real change violates a protected metric, invariant, slice, or approved envelope.	Block promotion and initiate attribution/bisection.
INCONCLUSIVE	Evidence is insufficient, unstable, or affected by uncontrolled environmental variation.	Do not equate with success; collect more evidence or freeze the environment.

Lane 1 - Current SSM functionality paired with adopted Auto ideas

This lane is the present v2.5.0-rc.1 product baseline. It should be documented as implemented capability, not as proposed research.

Current capability	Technical function	Research value
Semantic product compilation	Unstructured intent -> RequirementsIR -> AppFoundationPlan -> ArchitectureIR -> capability negotiation -> SML -> SIR -> deterministic target generation.	Creates stable intermediate representations that can be fingerprinted and compared by stage.
Capability honesty	Domain packs, supported/unsupported features, assumptions and negotiation status are explicit.	Prevents aggregate success from hiding capability loss or silent unsupported behaviour.
Compiler-owned target generation	Python, SQLAlchemy, Alembic, SaaS primitives, React admin and evidence artifacts are deterministic outputs.	Separates model/scaffolding variation from deterministic backend generation.
Bounded online drafting and repair	Models draft or repair SML within compiler diagnostics and retry limits; source generation remains compiler-owned.	Provides a controlled nondeterministic subsystem for repeated-run study.
Adopted Auto trust records	Generated app manifest, app contract, eval run, capability report, assumptions, unsupported features, provenance hashes, repair trace and evidence bundle.	Provides the first run-level evidence layer and release traceability.
Release gates and lock	Framework tests, generated-app tests, static analysis, migrations, frontend build, evidence validation, live-provider repair gate and immutable RC tag.	Supplies externally inspectable outcomes for candidate release comparison.

Immediate consolidation decision

Create one canonical GenerationRunRecord that links all existing SSM evidence artifacts with environment identity: compiler version, provider/model identifier, scaffold version, prompt version, dependency lock digest, benchmark intent ID, SML/SIR digests, repair attempts, generated test results, cost, latency and artifact hashes. This is schema consolidation, not a new compiler feature.

Acceptance checkpoint A: every benchmark generation run is reproducible or explicitly marked non-reproducible, and every metric used later by the research assay is either measured or absent - never inferred from an aspirational manifest.

Lane 2 - Additional Auto capabilities to integrate for research setup

Integrate only the Auto functions required to make SSM runs comparable, attributable and statistically analyzable. Full cognition-binary deployment is not required for the initial research proposal.

Priori ty	Auto capability	SSM integration	Why it is needed
P0	Trace SDK and replay comparison	Capture model calls, tool calls, state/context operations, arguments, outputs, timing, errors and digests during SML drafting/repair.	Creates per-run behavioural evidence rather than relying only on final pass/fail.
P0	Determinism census	Group equivalent call signatures and label them witnessed deterministic, divergent or unwitnessed.	Measures where variability originates and separates symbolic/stable spans from stochastic residue.
P0	Content-addressed eval runs	Persist immutable evaluation records keyed by contract, environment and observation digests.	Prevents evidence drift and supports exact baseline/candidate pairing.
P1	Behavioural contract + three-valued verification	Declare examples, properties, budgets and thresholds; produce Pass, Fail or Unchecked.	Turns acceptable behaviour into a versioned, machine-readable research object.
P1	Differential verifier	Compare candidate runs with a frozen reference interpreter or release under declared equivalence rules.	Supplies matched observations for paired tests and attribution.
P1	Registry and signing	Store manifests/eval records/artifacts by digest and optionally sign release evidence.	Makes research results auditable and suitable for software-lock evidence.
P2	Conformal guard, deopt and ratchet	Use only after the offline assay is stable; route novel cases to the frontier path and capture them for recompilation.	Supports later production monitoring and online adaptation studies.
Deferred	WASM cognition binary and zero-import confinement	Evaluate as a separate product/research extension, not a prerequisite for the first study.	Avoids expanding the proposal into a runtime-replacement thesis.

Engineering sequence

Checkpoint	Deliverable	Exit criterion
B1 - Observation	Trace schema, recorder, replay tool and environment lock.	Same run can be replayed or its non-reproducible dependencies are explicit.
B2 - Contract	Behavioural contract schema and content-addressed eval record.	Every metric has a source observation and declared acceptance rule.
B3 - Corpus	SSM-Bench v1 with 30-50 matched intents across CRUD, workflows, tenancy, RBAC, temporal fields, PATCH invariants and contextual rules.	Corpus is frozen before baseline data collection.

Lane 3 - Research proposal integration and possible extensions

The research contribution should be framed above Auto, not as a duplicate of Auto. Auto measures and verifies individual artifacts; the research layer performs longitudinal statistical inference across evolving SSM releases and production traces.

Research stage	Core question	Method and output
R1 - Offline noise floor	How much does the same frozen SSM release vary when nothing changes?	Repeated runs on matched SSM-Bench intents; per-metric distributions; environment variance audit.
R2 - Release comparison	Did candidate release N+1 materially change from baseline N?	Paired tests on per-intent outcomes, effect sizes, confidence bounds and false-alarm control.
R3 - Intended evolution	Is a real change authorised or accidental?	Versioned ChangeIntentContract with objectives, protected invariants, accepted trade-offs and affected slices.
R4 - Attribution	Where did the first material divergence occur?	Stage fingerprints across RequirementsIR, ArchitectureIR, SML, SIR, repair trace and generated artifacts; commit/scaffold bisection.
R5 - Hidden slices	Is an aggregate pass rate concealing a regression in one capability or architecture family?	Slice-aware tests over domain pack, backend, tenancy, workflow, update-model and rule-complexity dimensions.

Possible proposal extensions

Extension	Research value	Dependency
Continuous assay in production	Sequential, anytime-valid monitoring of behavioural fingerprints without repeated identical requests.	Stable offline assay, trace schema and baseline governance.
Closed or versionless provider drift	Detect change behind a fixed API/model name using environment and output-distribution evidence.	Provider-independent fingerprints and sequential tests.
Unknown-slice discovery	Find emerging subpopulations where regressions hide without pre-labelled traffic categories.	Sufficient telemetry, clustering/partition controls and multiplicity correction.
Cost-latency-quality trade-off certification	Classify a decline as intended only when measured effects stay inside an approved multi-objective envelope.	ChangeIntentContract and protected-metric policy.
Auto guard/ratchet study	Test whether novelty guards and recompilation preserve contract behaviour during distribution shifts.	P2 Auto runtime integration and production-like streams.

Novelty boundary: the proposal is not "detect whether an agent is nondeterministic" and not "verify one artifact against a contract." It is specification- and provenance-conditioned statistical detection, classification and attribution of behavioural change across releases and live traces.

Production decisions, governance and next milestone

Decision	Production rule
Separation of planes	SSM owns product semantics and deterministic generation; Auto owns observation/reproducibility contracts; the assay owns statistical release judgement.
Measured-only evidence	A manifest may contain a measured value or null. No research or release claim may be reconstructed from an undocumented assumption.
Baseline governance	Baselines are immutable and content-addressed. A monitor cannot silently update its own acceptable distribution.
Explicit change intent	Intended evolution must be declared before candidate evaluation, including protected metrics and accepted trade-offs.
Four-state decision	INCONCLUSIVE is a first-class result and must not be collapsed into NO MATERIAL CHANGE.
Fail-closed integration	Missing required evidence, contract failure, evidence hash mismatch or uncontrolled environment drift blocks certification.
No automatic semantic repair	A statistical finding may trigger diagnosis or a new engineering branch; it may not directly mutate compiler rules or generated source.

Minimum viable integrated study

Work package	Concrete output
WP1 - Canonical run record	GenerationRunRecord schema linking SSM manifests, Auto-style trace/eval IDs and environment locks.
WP2 - SSM-Bench v1	Frozen 30-50 intent corpus with stratified capability and complexity slices.
WP3 - Repeated baseline	10-20 runs of one locked SSM release under one locked provider/model environment.
WP4 - Candidate perturbations	At least one no-change control, one known regression, and one declared cost/latency trade-off.
WP5 - Statistical assay	Paired release comparison with effect sizes, false-alarm control and four-state verdict.
WP6 - Attribution report	First-changed-stage analysis using IR, trace, repair and generated-artifact fingerprints.

Recommended next checkpoint: implement WP1-WP2 only, freeze their schemas and corpus, then review the research question and statistical design before collecting expensive repeated-run data.

Source basis

This plan is grounded in the locked SSM Schrodinger v2.5.0-rc.1 product documentation and release evidence; the uploaded Auto source and README describing trace compilation, typed IR, behavioural contracts, content-addressed eval runs, measured manifests, guards and the tier-0/tier-1 ratchet; and the uploaded research draft "From Why Did My Agent Get Worse to a PhD," particularly its distinction between regression, intended change and ordinary variation, and its offline-to-production research arc.